Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

JamiHyari D. Andaya
BSIT-4A

PROFESSIONAL INSTALLATION MANUAL

Ubuntu Server 24.04.4 LTS Deployment

1. Introduction

Ubuntu Server is a Linux-based operating system widely used for web hosting, database services, file sharing, cloud environments, containerized applications, and other enterprise services. Unlike Ubuntu Desktop, Ubuntu Server typically runs without a graphical user interface, allowing administrators to manage the system through terminal commands, configuration files, and secure remote access.

This manual provides a complete guide for installing and configuring Ubuntu Server 24.04.4 LTS using Oracle VirtualBox. It includes preparing the virtual machine, installing the operating system, performing the initial configuration, verifying server functionality, troubleshooting common issues, and following recommended administrative practices. The guide is designed for system administrators and students who need to replicate a working Ubuntu Server environment.

After completing the deployment, the server will have a properly configured hostname, an administrative user account, network connectivity, updated system packages, and an operational OpenSSH service.

2. System Requirements

Before beginning the installation, prepare the following:

- Oracle VirtualBox 7.2 or a compatible version
- Ubuntu Server 24.04.4 LTS AMD64 ISO
- At least 4 GB of available host memory
- At least 40 GB of available storage
- Internet connection
- A computer with hardware virtualization enabled

The correct installation image should have a filename similar to:

- ubuntu-24.04.4-live-server-amd64.iso

Do not use an ISO containing desktop-amd64, because that file installs Ubuntu Desktop instead of Ubuntu Server.

3. Virtual Machine Specifications

Create a new virtual machine using the following settings:

Component	Recommended Configuration
Name	Ubuntu-Server-Week03
Type	Linux
Version	Ubuntu 64-bit
Memory	4096 MB
Processors	2
Virtual Disk	40 GB VDI
Disk Allocation	Dynamically allocated
Network Adapter	NAT
Optical Drive	Ubuntu Server ISO
Hostname	server01

Ensure that the Ubuntu Server ISO is attached to the virtual optical drive before starting the machine.

4. Ubuntu Server Installation Guide

4.1 Start the Virtual Machine

Start Ubuntu-Server-Week03 in Oracle VirtualBox. When the Ubuntu boot menu appears, select:

- Try or Install Ubuntu Server

The standard option is recommended for a virtual machine. The HWE kernel is normally unnecessary unless newer hardware support is specifically required.

4.2 Select the Language

Select:

- English

If the installer offers an update, select:

- Continue without updating

The installed operating system will be updated after deployment.

4.3 Configure the Keyboard

Use the following settings:

- Layout: English (US)
- Variant: English (US)

Select **Done** to continue.

4.4 Select the Installation Type

Use the standard Ubuntu Server installation. Do not select a minimized installation unless specifically required.

Continue without loading additional third-party drivers unless the virtual machine requires them.

4.5 Configure the Network

The VirtualBox network adapter should appear as an interface such as:

- Enp0s3

Leave the interface configured for DHCP. The system should automatically receive an IP address from VirtualBox NAT

Do not create a network bond because only one virtual network adapter is being used.

Select **Done** when an IP address appears.

4.6 Configure the Proxy

Leave the proxy address blank unless an organization has provided a specific proxy server.

Select **Done**.

4.7 Confirm the Ubuntu Archive Mirror

Keep the default Ubuntu archive mirror. Wait for the mirror test to complete successfully, then select **Done**.

4.8 Configure Storage

Select:

- Use an entire disk

Choose the 40 GB VirtualBox disk. Keep the option to configure the disk as an LVM group enabled. Leave disk encryption disabled unless it is required by organizational policy.

Review the proposed storage layout. It should contain the required boot partition, ext4 file system, and logical volume mounted at /.

Select **Done**, then confirm the destructive action by selecting Continue. This action formats only the selected virtual disk, not the host computer's physical storage.

4.G Configure the Server Profile

Enter the following information:

- Your name: Nico Adrielle Montorio
- Your server's name: server01
- Username: nico
- Password: A strong private password

Use a password containing uppercase and lowercase letters, numbers, and special characters. Do not include spaces in the username or hostname.

Store the password securely because it is required for login and sudo commands

4.10 Ubuntu Pro

Select:

- Skip for now

Ubuntu Pro is optional and is not required for a basic laboratory deployment.

4.11 SSH Configuration

Enable:

- Install OpenSSH server

Do not import an SSH identity unless a public key has already been prepared.

OpenSSH allows the server to be managed securely from another computer.

4.12 Featured Server Snaps

Do not select additional server snaps unless they are specifically required. Continue by selecting **Done**.

4.13 Complete the Installation

Wait while the installer copies and configures the operating-system files. When installation finishes, select:

- Reboot Now

If the following message appears:

  - Please remove the installation medium, then press ENTER

Open the VirtualBox **Devices** menu, remove the Ubuntu ISO from the virtual optical drive, and press Enter.

If the ISO is already removed, press Enter to continue. The virtual machine should restart from the installed virtual hard disk.

5. Initial Server Configuration

5.1 Log In

At the login prompt, enter:

- Username: nico
- Password: The password created during installation

Linux does not display dots or asterisks while a password is entered. This behavior is normal.

A successful login displays a prompt similar to:

nico@server01:~$

5.2 Update the Server

Refresh the package information:

- sudo apt update

Install all available updates:

- sudo apt upgrade –y

Restart the server if the update process installs a new kernel or indicates that a reboot is required:

- sudo reboot

5.3 Install OpenSSH Server If Necessary

If OpenSSH was not enabled during installation, install it manually:

- If OpenSSH was not enabled during installation, install it manually:

Start the service:

- sudo service ssh start

Enable SSH to start automatically during boot:

- sudo systemctl enable ssh

5.4 Check the Firewall

Check the firewall status:

- sudo ufw status

If the firewall will be enabled, allow SSH first:

- sudo ufw allow OpenSSH

Enable the firewall:

- sudo ufw enable

Confirm its rules:

- sudo ufw status

Allowing OpenSSH before enabling the firewall prevents remote administrators from being locked out.

6. Server Verification Procedures

6.1 Verify the Hostname

Run:

- Hostname

Expected result:

- Server01

Additional system information can be viewed using:

- Hostnamectl

6.2 Verify the IP Address

Run:

- ip addr

Locate the active interface, normally enp0s3, and identify the line beginning with inet.

A VirtualBox NAT address may appear as:

- 10.0.2.15/24

The exact address may differ because DHCP assigns it automatically.

6.3 Test Internet Connectivity

Test both network and DNS connectivity:

- ping -c 4 google.com

A successful test returns four replies and reports zero packet loss or another acceptable result.

If domain-name resolution fails, test a public IP address:

- ping -c 4 8.8.8.8

If the IP address works but the domain name fails, investigate the DNS configuration.

6.4 Verify System Updates

Run:

- sudo service ssh status

The service should display:

- Active: active (running)

Press Q to exit the status display.

The following command may also be used:

- The following command may also be used:

6.6 Verify Disk Usage

Run:

- df –h

Confirm that the root file system mounted at / has sufficient available storage.

6.7 Verify Memory and Processor Information

Display memory usage:

- free –h

Display processor information:

- Lscpu

Confirm that the virtual machine recognizes the configured memory and two virtual processors.

7. Troubleshooting Guide

7.1 The Installer Shows Ubuntu Desktop

Cause: The Ubuntu Desktop ISO was attached instead of Ubuntu Server

Solution: Shut down the VM and attach an ISO whose filename contains:

- live-server-amd64.iso

Restart the virtual machine and select **Try or Install Ubuntu Server**.

7.2 The Installer Freezes During Startup

Possible causes:

- Incorrect or corrupted ISO
- Insufficient virtual-machine resources
- Virtualization conflict
- Incorrect VirtualBox settings

Solutions:

- Confirm that the correct Ubuntu Server AMD64 ISO is being used.
- Assign 4096 MB of memory and two processors.

- Verify that hardware virtualization is enabled.
- Restart the virtual machine.
- Verify the downloaded ISO checksum if the problem continues.

7.3 Network Interface Has No IP Address

Check that the VirtualBox network adapter is enabled and attached to NAT.

Restart the network service:

- sudo systemctl restart systemd-networkd

Check the interface again:

- ip addr

If necessary, restart the virtual machine:

- sudo reboot

7.4 Internet Ping Fails

Check the network interface:

- ip addr

Test the VirtualBox NAT gateway:

- ping -c 4 10.0.2.2

Test a public IP address:

- ping -c 4 8.8.8.8

Test domain-name resolution:

- ping -c 4 google.com

These tests help determine whether the issue involves the local interface, internet connection, or DNS.

7.5 SSH Service Cannot Be Found

If the system reports:

- Unit ssh.service could not be found

Install OpenSSH Server:

- sudo apt update
- sudo apt install openssh-server –y

Start and enable it:

- sudo systemctl enable --now ssh

Verify its status:

- systemctl status ssh

7.6 Incorrect Systemclt Command

The correct spelling is:

- systemctl

It consists of system followed by ctl. There must be no space between them, and the final character is a lowercase letter l.

If needed, use this alternative command:

- sudo service ssh status

7.7 Installation Media Cannot Be Unmounted

If Ubuntu reports that /cdrom could not be unmounted, remove the ISO through:

VirtualBox → Devices → Optical Drives → Remove disk from virtual drive

Press **Enter** afterward. If the ISO is already removed, power off the completed VM, confirm that the optical drive is empty, and start the VM again.

7.8 Login Is Incorrect

Confirm that the correct lowercase username is being used. Password characters are invisible while typing.

Check:

- Caps Lock is disabled.
- The correct keyboard layout is active
- The password is entered exactly as created.

Avoid repeatedly guessing passwords. Use recovery procedures if access cannot be restored.

7.G The Server Boots Back Into the Installer

The installation ISO is still mounted or the optical drive appears before the hard disk in the boot order

Remove the ISO from the virtual optical drive and restart the virtual machine.

8. Best Practices

8.1 Use Strong Passwords

Administrative accounts must use unique passwords containing a combination of uppercase letters, lowercase letters, numbers, and special characters. Passwords must not be included in screenshots or documentation.

8.2 Use a Non-Root Account

Perform normal administration using a non-root user with sudo privileges. Direct root login should be avoided because it increases the impact of mistakes and unauthorized access.

8.3 Keep the Server Updated

Regularly run:

- sudo apt update
- sudo apt upgrade

Security updates should be applied promptly. Reboot the server when a kernel or critical system update requires it.

8.4 Secure SSH Access

Use SSH keys instead of passwords for production remote access. Disable direct root login and consider disabling password authentication after confirming that keybased authentication works.

The SSH configuration file is located at:

- /etc/ssh/sshd_config

Back up the file before modifying it and test configuration changes before ending an active remote session.

8.5 Enable Only Required Services

Every unnecessary service increases resource consumption and potential security exposure. Install only the packages and server roles required for the intended workload.

List running services using:

- systemctl --type=service --state=running

8.6 Configure the Firewall

Allow only required ports. Use UFW to review and manage firewall rules:

- sudo ufw status

For a basic remote server, SSH access may be permitted while unrelated inbound services remain blocked.

8.7 Monitor Storage and Resources

Regularly check disk usage:

- df –h

Check memory:

- free –h

Check active processes:

- top

Resource monitoring helps prevent performance problems and service interruptions.

8.8 Create Backups and VirtualBox Snapshots

Create a snapshot after completing a stable installation and initial configuration. Snapshots are useful before risky laboratory changes, but they should not replace proper backups of important files and configuration data.

Use clear snapshot names, such as:

- Clean Installation
- SSH Configured
- Before Web Server Setup

8.G Maintain Documentation

Record the following information:

- Hostname and IP address
- Operating-system version
- Installed services
- Firewall rules
- Configuration changes
- Update history
- Backup procedures
- Known problems and solutions

Never record passwords, private keys, or other authentication secrets in general documentation.

8.10 Shut Down the Server Properly

Use:

- sudo shutdown now

or:

- sudo poweroff

Avoid closing or powering off the virtual machine while the system is writing data unless it is completely unresponsive.

G. References

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